

PERFECT LINEAR CODES, COMPLETE PRIME POWERED ORTHOGONAL ARRAYS AND SATURATED PRIME POWERED FRACTIONAL FACTORIAL DESIGNS

H. Pesotan* B.L. Raktoe† & D.O. Chalton†

*Department of Mathematics and Statistics, University of Guelph, Guelph, Ontario, Canada, N1G 2W1.

E-mail address: hpesotan@msnet.mathstat.uoguelph.ca.

†Department of Statistics, University of the Western Cape, Private Bag X17, Bellville, 7535, South Africa.

E-mail address: dchalton@uwc.ac.za

Key words: Linear codes; orthogonal arrays; saturated designs.

Summary: This paper discusses the concepts of perfect prime powered linear codes, complete prime powered orthogonal arrays and saturated prime powered fractional factorial designs. It is established that the existence of a perfect prime powered linear code implies both the existence of a complete prime powered orthogonal array and the existence of a saturated prime powered fractional factorial design. In the development we provide a known comprehensive list of perfect prime powered linear codes gleaned from the literature.

1. Introduction

It is at present quite common to see problems and their solutions being studied independently in several algebraic-geometric-combinatorial settings. Once the definitions, terminologies and notations in the various settings are unmasked it

turns out that the problems and solutions are equivalent. This paper has the objective to show that this sort of parallel development need not be so.

In the present paper three settings are considered. These are (i) perfect prime powered linear codes, (ii) complete prime powered orthogonal arrays, and (iii) saturated prime powered fractional factorial designs. Specifically, it is established that the existence of perfect error correcting $[k, p, d]$ -linear codes implies the existence of certain complete prime powered OA (n, m, s, t) orthogonal arrays and certain saturated prime powered s_R^{k-p} fractional factorial designs.

The results in this paper have practical applicability in an experimental design setting. When k factors, each at s levels, are studied in an experiment, one possibility is to select the complete s^k factorial to conduct the investigation. However, such a choice involves a large number of runs which may be too expensive to conduct and unnecessary. Thus when the size of a design is of concern, a fractional factorial design (FFD) is often used. A good choice of such a design allows for the study of many factors with a relatively small number of runs and yet allows for the estimation of a large number of effects. In this context, when s is a prime or prime power, regular FFD's are often used. Such fractions are obtained from defining relations and are examples of orthogonal arrays. Orthogonal arrays, however, are a much more extensive class of designs and were introduced by Rao (1947). FFD's with each factor at two levels are often used in factor screening experiments when not much is known about how the factors affect a response of interest. Orthogonal arrays with three levels are useful when linear and quadratic effects of a factor need to be tested or detected. Thus the construction of regular FFD's of a given resolution and which are optimal in the sense that they have

the fewest number of runs is a worthy endeavour. In this paper we will present, when s is a prime or prime power, certain optimal regular FFD's and complete prime powered orthogonal arrays via the use of coding theory.

In Section 2 the concepts, definitions, problems and solutions of perfect linear $[k, p, d]$ -codes are presented together with the exhaustive list of six such codes. Two of these are trivial perfect linear codes and the remaining four are non-trivial perfect linear codes. These four we will call the useful perfect linear codes (upc's). These four upc's then lead to four complete prime powered OA (n, m, s, t) orthogonal arrays in Section 3, and, to four saturated prime powered s_R^{k-p} fractional factorial designs in Section 4. In Section 5 a discussion is provided to encourage the study of the theories and results in these parallel areas.

The literature in the areas of codes, orthogonal arrays and factorial designs is enormous. For readers interested in these areas we provide some selected references apart from those mentioned in the development of the material in the paper. These are: Box and Hunter (1961), Constantine (1987), Dey (1985), Draper and Mitchell (1967), Federer and Mandeli (1986), Hedayat, Stufken and Su (1997), Hedayat et al. (1999), Raktoc et al. (1980) and Street and Street (1987).

2. Perfect linear codes

Let $\text{GF}(s)$ be the Galois field of order s where s is a prime or prime power and let $V(k, s)$ be the vector space of dimension k over $\text{GF}(s)$. The finite field $\text{GF}(s)$ is the *code alphabet* and it has an element denoted by 0 which is the additive identity of the field. A vector x in $V(k, s)$, written as a row vector, is called a *codeword*. An s -ary linear code (or simply *linear*

code) L is a subspace of $V(k, s)$. The *Hamming distance* $d(x, y)$ of two codewords x and y in $V(k, s)$ is the number of positions in which x and y differ. The *weight* $w(x)$ of a codeword x is $d(x, 0)$, and is thus the number of non-zero coordinates in x . The *minimum weight* $w(L)$ of a linear code L is the minimum weight of all non-zero code words in L . If a linear code has dimension p over $V(k, s)$ we say that L is a $[k, p]$ -code, and if L has minimum weight d it is called a $[k, p, d]$ -code. In the literature on coding theory the symbols k, p and d are referred to as *length, dimension* and *minimum distance*, respectively. Since both of these codes have s^p codewords they are also called a (k, s^p) -code and (k, s^p, d) -code respectively. We say that a $[k, p, d]$ -code is *e-error correcting* if $d \geq 2e + 1$ and its error correcting capability is $e = \left\lfloor \frac{d-1}{2} \right\rfloor$, where $\lfloor \cdot \rfloor$ denotes the greatest integer function.

Let L be a $[k, p]$ -code. Then a $p \times k$ matrix G whose rows form a basis for L is called a *generator matrix*. For a $[k, p]$ -code the generator matrix can be brought by elementary row operations into the *standard form* $G = (I_p | A)$, where I_p is the identity matrix of order p and A is a $p \times (k-p)$ matrix. Thus $L = \{xG : x \in V(p, s)\}$. If L is a $[k, p]$ -code, then the set $L^\perp = \{x \in V(k, s) : x \cdot c = 0, \forall c \in L\}$ is called the *dual code* of L . Here $x \cdot y$ is the usual *dot product* for two vectors x and y in $V(k, s)$, i.e. $x \cdot y = \sum_{i=1}^k x_i y_i$. It can be shown that if G is the generator matrix for L then $L^\perp = \{x \in V(k, s) : xG' = 0\}$. Also, the dual $L^\perp$ of a $[k, p]$ -code is a $[k, k-p]$ -code. Further, it is easily verified that $L^{\perp\perp} = L$. A linear code L for which $L = L^\perp$ is said to be *self-dual*. If L is a $[k, p]$ -code with standard generator matrix $G = (I_p | A)$ then the matrix $H = (-A' | I_{k-p})$ is called the *parity check matrix* for L with the property that $GH' = O$. It

follows that $x \in L$ if and only if $xH' = 0$. The parity check matrix H is a *generator matrix* for the dual code $L^\perp$ and is not in standard form.

Now let L be a $[k, p, d]$ -code with parity check matrix H . It can then be shown that d is the smallest integer r for which there are r linearly dependent columns in H . Thus, H has d linearly dependent columns, but any $d - 1$ columns are linearly independent. This result allows one to obtain the minimum weight d of a linear code from the parity check matrix H since it is difficult to obtain it from the generator matrix G .

Let L be a $[k, p, d]$ -code with *packing radius* $e = \left\lceil \frac{d-1}{2} \right\rceil$. The set $S_e(x) = \{x \in V(k, s) : w(x) \leq e\}$ is called the *sphere of radius e* around x . If the set of spheres $S_e(y)$, where y runs through the code L , forms a partition of $V(k, s)$ then L is called a *perfect code*.

The *optimum coding problem* for a $[k, p, d]$ -code L is to find the maximum value $p_{\max}(k, d)$ of p for given k and d , where $p_{\max}(k, d) = \max\{p: \text{there exists a } [k, p, d]\text{-code}\}$. On the other hand the *optimum packing problem* for a $[k, p, d]$ -code L is to find the maximum value $k_{\max}(p, d)$ of k for given p and d , where $k_{\max}(p, d) = \max\{k: \text{there exists a } [k, p, d]\text{-code}\}$.

Let $L^\perp$ be the dual $[k, q = k - p]$ -code corresponding to the $[k, p, d]$ -code L . It is clear that maximization of the dimension p of the code L is equivalent to the minimization of the codimension q of $L^\perp$. The optimal coding problem in terms of q is to find the minimum value $q_{\min}(k, d)$ of q for given k and d , where $q_{\min}(k, d) = k - p_{\max}(k, d)$. Also, the packing problem in terms of q is to find the maximum value $k_{\max}(q, d)$ of k , where $k_{\max}(q, d) = k_{\max}(p, d)$.

It is well known that the optimum coding problem and the optimum packing problem are equivalent in the following sense:

$$q_{\min}(k, d) \leq q \iff k_{\max}(q, d) \geq k. \quad (2.1)$$

Further:

$$\begin{cases} q_{\min}(k, d) = \min \{q : k_{\max}(q, d) \geq k\}, \text{ and} \\ k_{\max}(q, d) = \max \{k : q_{\min}(k, d) = q\}. \end{cases} \quad (2.2)$$

The Sphere-Packing Bound or Hamming Lower Bound for $q_{\min}(k, d)$ is given by:

$$q_{\min}(k, d) \geq \log_s \left(\sum_{i=0}^e \binom{k}{i} (s-1)^i \right), \quad e = \left\lfloor \frac{d-1}{2} \right\rfloor, \quad (2.3)$$

and the Rao-Hamming Lower Bounds are as follows:

$$\begin{cases} q_{\min}(k, d = 2e + 1) \geq \log_s \left(\sum_{i=0}^e \binom{k}{i} (s-1)^i \right), \text{ and} \\ q_{\min}(k, d = 2(e + 1)) \geq \log_s \left(\sum_{i=0}^e \binom{k}{i} (s-1)^i + \binom{k-1}{e} (s-1)^{e+1} \right). \end{cases} \quad (2.4)$$

Note that for $d = 2e + 1$ the bounds in (2.3) and (2.4) are identical.

Both the packing and coding problems have been solved for $d = 3$ and for $d \geq 4$ they remain largely unsolved. There are tables available for many values of s and the other two parameters k and d e.g. see Verhoeff (1987). For $d = 3$ it is easy to show that $q_{\min}(k, 3) = q \iff k_{\max}(q, 3) = \frac{s^q - 1}{s - 1}$.

The following two results are pertinent to perfect linear codes and can be found in standard books on coding theory, e.g. see Conway and Sloane (1993), Roman (1992) and Van Lint (1977).

THEOREM 2.1. Let L be a $[k, p, d]$ -code. Then L is perfect if and only if $d = 2e + 1$ is odd and equality holds in the Sphere-Packing Bound or

the Hamming Bound, i.e.

$$q_{\min}(k, d) = \log_s \left(\sum_{i=0}^e \binom{k}{i} (s-1)^i \right), \quad e = \frac{d-1}{2}. \quad (2.5)$$

THEOREM 2.2. The following $[k, p, d]$ -codes are the unique (in the usual sense) perfect linear codes and represent an exhaustive list in our prime powered setting:

- (i) the $[k, 0, 2k+1]$ -code consisting of the codeword 0 with d being undefined;
- (ii) the $[k, k, 1]$ -code consisting of the entire vector space $V[k, s]$;
- (iii) the binary (i.e. $s = 2$) code $B_{2e+1} = [2e+1, 1, 2e+1]$ -code, which is the linear repetition code of odd length $k = 2e+1$;
- (iv) the Hamming code $H_s(r) = \left[\frac{s^r-1}{s-1}, k-r, 3 \right]$ -code of order r ,
 where $k = \frac{s^r-1}{s-1}$;
- (v) the binary (i.e. $s = 2$) Golay code $G_{23} = [23, 12, 7]$ -code; and
- (vi) the ternary (i.e. $s = 3$) Golay code $G_{11} = [11, 6, 5]$ -code.

In the literature on coding theory the first two codes are known as *trivial perfect codes* and the remaining ones are called *non-trivial perfect codes*. It will be seen in subsequent sections that only codes (iii) to (vi) will have real meaning for our purposes and for this reason we will refer to these four codes as *useful perfect codes* (upc's). These four upc's are optimal codes both from the coding problem and the packing problem viewpoints. Specifically note that for these codes we have:

$$\begin{aligned} B_{2e+1} : q_{\min}(2e+1, 2e+1) = 2e &\iff k_{\max}(2e, 2e+1) = 2e+1, \\ H_s(r) : q_{\min}\left(\frac{s^r-1}{s-1}, 3\right) = r &\iff k_{\max}(s^r, 3) = \frac{s^r-1}{s-1}, \\ G_{23} : q_{\min}(23, 7) = 11 &\iff k_{\max}(11, 7) = 23, \text{ and} \\ G_{11} : q_{\min}(11, 5) = 5 &\iff k_{\max}(5, 5) = 11. \end{aligned}$$

REMARK 3.1. Noda (1979) characterizes all sequences $(n, m, s, 4)$ which satisfy Rao's bound (3.1). These sequences include the complete arrays $OA[2^4, 5, 2, 4]$ and $OA[3^5, 11, 3, 4]$ whose constructions are well known in coding theory. Kageyama (1988) characterizes all sequences $(n, m, 2, t)$ which satisfy Rao's bound (3.1). These sequences include the complete arrays $OA[2^{2e}, 2e+1, 2, 2e]$ and $OA\left[s^r, \frac{s^r-1}{s-1}, s, 2\right]$. Both of these authors do not mention perfect linear codes, and yet these have had a long history (e.g. see Van Lint (1977) and the references mentioned by him). Apart from settling that all the four OA 's mentioned in the above theorem are complete, it is established for the first time that the $OA[2^{2e}, 2e+1, 2, 2e]$ and $OA[2^{11}, 23, 2, 6]$ are complete, and this was accomplished via the perfect code route.

REMARK 3.2. It is easy to check that the optimal self-dual codes $G_{24} = [24, 12, 8]$ -code and $G_{12} = [12, 6, 6]$ -code are the complete orthogonal arrays $OA[2^{12}, 24, 2, 7]$ and $OA[3^6, 12, 3, 5]$ respectively. The completeness of G_{12} is mentioned by Kageyama (1988) but not that of G_{24} and so this is something new.

4. Saturated prime powered fractional factorial designs

Consider a *symmetrical factorial design* with k factors, each at s levels, where s is a prime or prime power. The set of s levels of each factor is made to correspond to the Galois field $GF(s)$ consisting of s elements in which there is an element identified as 0 and is the additive identity of the field. The k -dimensional vector space $V(k, s)$ over $GF(s)$ represents the *complete factorial design* consisting of s^k treatment combinations. A

fractional factorial design is said to be a *regular design* if it is a $(k - p)$ -dimensional subspace of $V(k, s)$ or a coset of such a subspace. Such a design consists of s^{k-p} treatment combinations and is a $\frac{1}{s^p}$ fraction of the complete factorial design. A *regular fractional factorial design of resolution R* is called an s_R^{k-p} design, where $3 \leq R \leq k$. In statistical terms an s_R^{k-p} design is such that all main effects are confounded (or aliased) with interactions involving $(R - 1)$ or more factors, and all two-factor interactions are confounded with interactions involving $(R - 2)$ or more factors, and so on. For details concerning factorial design the reader may choose among the references made in Section 1. Moreover, the book by Raktue, Hedayat and Federer (1981) contains extensive references on factorial design articles prior to 1981.

Consider an experimental situation in which we seek a *FFD* D with N runs from the complete s^k factorial which is capable of estimating: (i) when R is odd: all factorial effects up to and including order $[R/2]$ under the assumption that all factorial effects of order $[R/2] + 1$ or higher are zero, where $[]$ denotes the greatest integer function, (ii) when R is even: all factorial effects up to and including order $(R/2) - 1$ under the assumption that all factorial effects of order $(R/2) + 1$ or higher are zero. Under these circumstances the questions are (a) what should be chosen for D ? and (b) what should be an appropriate value of N ? The choice of D is usually taken to be a regular fraction of resolution R with $N = s^q$ runs. Construction of designs of this kind has been studied (see, e.g., Raktue, Hedayat and Federer (1981) and the references therein). To make the experiment cost efficient one needs to economize on the number of runs $N = s^q$ and thus the problem is to find the least value q for which there exists a regular *FFD* D of resolution

R with s^q runs. Hence, from the viewpoint of applications, construction of regular s_R^{k-p} designs with the minimum value for $q = k - p$ is desirable. In this section we will construct such designs via the perfect code route.

For an s_R^{k-p} design there are two design problems which are similar to the coding and packing problems of Section 2. Specifically, these are:

- (i) Minimize the fraction $\frac{1}{s^p}$ for a given number k of factors and a given resolution R (i.e. maximize p for given k and R) or equivalently minimize the number s^{k-p} of experimental units for a given number k of factors and a given resolution R (i.e. minimize $k - p$ for given k and R),
- (ii) Maximize the number k of factors for a given number s^{k-p} of experimental units and a given resolution R (i.e. maximize k for given $k - p$ and R).

To discuss these two problems from the linear coding theory viewpoint, it is sufficient to take an s_R^{k-p} as a $(k - p)$ -dimensional subspace of $V(k, s)$. Denote the minimum value of $(k - p)$ for given k and R by $(k - p)_{\min}(k, R)$ and the maximum value of k for given $k - p$ and R by $k_{\max}(k - p, R)$. Explicitly $(k - p)_{\min}(k, R) = \min\{k - p: \text{there exists an } s_R^{k-p} \text{ design}\}$ and $k_{\max}(k - p, R) = \max\{k: \text{there exists an } s_R^{k-p} \text{ design}\}$. Bose (1961) showed the equivalency of the two problems above in the sense that

$$(k - p)_{\min}(k, R) \leq k - p \iff k_{\max}(k - p, R) \geq k. \quad (4.1)$$

Furthermore,

$$\begin{cases} (k - p)_{\min}(k, R) = \min\{k - p: k_{\max}(k - p, R) \geq k\}, \text{ and} \\ k_{\max}(k - p, R) = \max\{k: (k - p)_{\min}(k, R) = k - p\}. \end{cases} \quad (4.2)$$

Lower and upper bounds for $(k - p)_{\min}(k, R)$, as mentioned in Raktue and Pesotan (1996) are given in (4.3) and (4.4) below.

$$\left\{ \begin{array}{l} (k - p)_{\min}(k, R = 2v + 1) \geq \{\log_s H(k, v)\}, \text{ and} \\ (k - p)_{\min}(k, R = 2(v + 1)) \geq \left\{ \log_s \left(H(k, v) + \binom{k-1}{v} (s-1)^{v+1} \right) \right\}, \\ \text{where } H(k, u) = \sum_{i=1}^u \binom{k}{i} (s-1)^i. \end{array} \right. \quad (4.3)$$

Moreover,

$$\left\{ \begin{array}{l} (k - p)_{\min}(k, R = 2v + 1) \leq 2v \{\log_s(k + 1)\}, \text{ and} \\ (k - p)_{\min}(k, R = 2(v + 1)) \leq (2v + 2) \{\log_s(k + 1)\} - 1. \end{array} \right. \quad (4.4)$$

Further, observe that (4.3) implicitly provide upper bounds and (4.4) lower bounds for $k_{\max}(k - p, R = 2v + 1)$ and $k_{\max}(k - p, R = 2(v + 1))$ respectively. An s_R^{k-p} design with the minimal exponent

$(k - p)_{\min}(k, R) = k - p$ (or equivalently with the maximum exponent $k - (k - p)_{\min}(k, R)$) will be called a *saturated design of resolution R in s^{k-p} treatment combinations*, or simply a *saturated s_R^{k-p} design*. We now can establish the following result for these designs.

THEOREM 4.1. The dual codes $B_{2e+1}^\perp$, $H_s^\perp(r)$, $G_{23}^\perp$ and $G_{11}^\perp$ are unique saturated $2_{2e+1}^{(2e+1)-1}$, $3_3^{(\frac{s^r-1}{s-1}) - ((\frac{s^r-1}{s-1})-r)}$, 2_7^{23-12} and 3_5^{11-6} designs.

PROOF. It is easy to observe if C is a linear $[k, p, d]$ -code then the dual code $C^\perp$ is an s_R^{k-p} design with $R = d$. Hence it follows immediately from the perfectness of these four codes that the corresponding designs are saturated and the uniqueness of the designs follows from the uniqueness of the codes.

REMARK 4.1. It is easy to verify that the self-dual codes $G_{24} = [24, 12, 8]$ -code and $G_{12} = [12, 6, 6]$ -code are the saturated designs 2_8^{24-12} and 3_6^{12-6} respectively.

REMARK 4.2. Note that, apart from terminology and notation, the optimum coding problem and the saturated design problem are identical.

5. Discussion

Aside from Theorems 3.1 and 4.1 the following additional specific results are established via the perfect code route:

- (i) It is shown that (a) the orthogonal arrays $OA[2^4, 5, 2, 4]$ and $OA[3^5, 11, 3, 4]$ are complete, (b) the $OA[2^{2e}, 2e+1, 2, 2e]$ and $OA\left[s^r, \frac{s^r-1}{s-1}, s, 2\right]$ are complete, (c) the $OA[2^{11}, 23, 2, 6]$ is complete. The completeness of the OA 's in (a) was established earlier by Noda (1979) and those in (b) by Kageyama (1988) using the Rao bounds (3.1) and (3.2). The completeness of the OA in (c) is established here for the first time via coding theory.
- (ii) The optimal self-dual code $G_{24} = [24, 12, 8]$ -code is a complete orthogonal array $OA[2^{12}, 24, 2, 7]$ and the self-dual codes $G_{24} = [24, 12, 8]$ -code and $G_{12} = [12, 6, 6]$ -code are the saturated designs 2_8^{24-12} and 3_6^{12-6} , respectively.

The kind of designs obtained via the perfect code route are powerful and optimal tools in statistical design of experiments and are extremely useful in industrial and agricultural experimentation as well as other areas of experimentation.

Finally, this paper shows that attention to results in coding theory which are pertinent to fractional factorial designs is a fruitful way of constructing such

designs. Papers which follow this approach include Srivastava (1990) and Raktoc and Pesotan (1996, 1997).

Acknowledgement

We thank the referees whose comments and helpful suggestions resulted in an improved presentation.

Research supported by NSERC Grant No. OGP0008776.

References

- BOSE, R.C. (1961). On some connections between the design of experiments and information theory. *Bull. Inst. Internat. Statist.*, **34**, 4^e, 257–271.
- BOX, G.E.P. & HUNTER, J.S. (1961). The 2^{k-p} fractional factorial designs I. *Technometrics*, **3**, 311–351.
- CONSTANTINE, G.M. (1987). *Combinational theory and statistical design*. Wiley, New York.
- CONWAY, J.H. & SLOANE, N.J.A. (1993). *Sphere packings, lattices and groups*. Second edition. Springer-Verlag, New York.
- DEY, A. (1985). *Orthogonal fractional factorial designs*. Wiley Eastern Limited, New Delhi.
- DRAPER, N.R. & MITCHELL, T.J. (1967). The construction of saturated 2_R^{k-p} designs. *Ann. Math. Statist.*, **38**, 1110–1126.
- DRAPER, N.R. & LIN, K. (1990). Capacity considerations for two-level fractional factorial designs. *J. Statist. Plann. Inference*, **24**, 25–35.
- FEDERER, W.T. & MANDELI, J.P. (1986). Orthogonal F-rectangles, orthogonal arrays and codes. *J. Comb. Theory Ser. A.*, **43**, 149–164.
- HEDAYAT, A., STUFKEN, J. & SU, GUOQIN. (1997). On the construction and existence of orthogonal arrays with three levels and indexes 1 and 2. *The Annals of Statist.*, **25**, 2044–2053.
- HEDAYAT, A.S., SLOANE, N.J.A. & STUFKEN, J. (1999). *Orthogonal arrays: Theory and applications*. Springer-Verlag, New York.

- KAGEYAMA, S. (1988). On orthogonal arrays attaining Rao's bounds. *J. Statist. Plann. Inference*, **19**, 395–400.
- NODA, R. (1979). On orthogonal arrays of strength 4 achieving Rao's bounds. *J. London Math. Soc. (2)*, **19**, 385–390.
- RAKTOE, B.L., HEDAYAT, A. & FEDERER, W.T. (1981). *Factorial designs*. Wiley, New York.
- RAKTOE, B.L. & PESOTAN, H. (1996). The critical problem of saturated s_R^{k-p} fractional factorial designs. *Math. Meth. Statist.*, **5**, 244–252.
- RAKTOE, B.L. & PESOTAN, H. (1997). Note on Draper and Lin's article "Capacity considerations for two-level fractional factorial designs". *J. Statist. Plann. Inference*, **59**, 163–165.
- RAKTOE, B.L., PESOTAN, H. & FEDERER, W.T. (1980). On aliasing and generalized defining relationships of factorial arrangements. *Can. J. Statist.*, **8**, 65–77.
- RAO, C.R. (1947). Factorial experiments derivable from combinatorial arrangements of arrays. *Suppl. J. Roy. Statist. Soc.*, **9**, 128–139.
- ROMAN, S. (1992). *Coding and information theory*. Springer-Verlag, New York.
- STREET, A.P. & STREET, D.J. (1987). *Combinatorics of experimental design*. Oxford University Press, New York.
- SRIVASTAVA, J.N. (1990). Modern factorial design theory. In: S. Ghosh, ed., *Statistical design and analysis of industrial experiments*. Marcel Dekker, New York, 311–406.
- VAN LINT, J.H. (1977). Codes and designs. In: M. Aigner ed., *Higher combinatorics*. D. Reidel Publishing Company, Dordrecht-Holland, 241–256.
- VERHOEFF, T. (1987). An updated table of minimum distance bounds for binary linear codes. *IEEE Trans. Inform. Theory* **IT-33**, 665–680.